

Clinic Management System — Complete Reference Manual

Version: June 2025

Repositories: `clinic-app-laravel-backend` (Laravel 13) + `clinic-app-laravel-frontend` (Vue 3)

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1. Architecture Overview

Philosophy

- **Service-oriented backend:** Every business operation is a `CoreService` class with `prepare()` → `process()` pipeline.

- **Config-driven CRUD:** Frontend renders tables and forms from per-entity TypeScript config files without writing boilerplate.
- **Multi-guard JWT:** Three separate auth guards (`api` , `doctor` , `patient`) share a single `/login` endpoint via auto-detection.
- **DB::table() queries:** CRUD services use raw query builder, not Eloquent ORM, for performance and explicit control.

Repository Map

Repo	Stack	Path
------	-------	------

Backend	Laravel 13, PHP 8.3, MySQL, tymon/jwt-auth	D:\Zka\Documents\projects\clinic-laravel
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<code>api</code>	App\Models\Users	<code>users</code>	Admin/staff users with <code>role_id</code>
------------------	------------------	--------------------	---

View list	<code>view-{model}</code>
-----------	---------------------------

<code>DoUnifiedLogin</code>	null	<code>/login</code>	Unified login (admin/doctor/patient)
-----------------------------	------	---------------------	--------------------------------------

<code>id</code>	BIGINT PK	auto
-----------------	-----------	------

<code>username</code>	VARCHAR(50)	UNIQUE
-----------------------	-------------	--------

<code>updated_at</code>	TIMESTAMP
-------------------------	-----------

id	BIGINT PK	auto
----	-----------	------

description	TEXT	nullable
-------------	------	----------

id	BIGINT PK	auto
----	-----------	------

module	VARCHAR(100)	
--------	--------------	--

id	BIGINT PK	auto
----	-----------	------

doctors

Column	Type	Notes
--------	------	-------

id	BIGINT PK	auto
----	-----------	------

specialization	VARCHAR(100)	
----------------	--------------	--

id	BIGINT PK	auto
----	-----------	------

email	VARCHAR(100)	UNIQUE
-------	--------------	--------

id	BIGINT PK	auto
----	-----------	------

capacity	INT UNSIGNED	default 1
----------	--------------	-----------

id	BIGINT PK	auto
----	-----------	------

appointment_time	TIME
------------------	------

id	BIGINT PK	auto
----	-----------	------

Users	users.ts	users	Avatar+name, role select, active badge
-------	----------	-------	--

```
-- POST /login ----->
```

```
-- Query Users by email ----->
```

```
<-- user record -----|
```

```
-- Hash::check(password) |
```

```
→ match: return JWT
```

```
-- (if no match: try Doctors)
```

```
→ match: return JWT |
```

```
-- (if no match: try Patients)|
```

```
→ match: return JWT
```

```
<-- {token, role, data}-
```

```
-- Store token in
```

```
-- Route to portal
```

CRUD Create Flow

Frontend

Backend

Storage

```
{field1, photo:
```

```
-- Validate rules
```

```
-- Check UNIQUE constraints|
```

```
-- Apply default values
```

```
-- beforeInsert() hook    |
```

```
-- Detect temp/ prefix
```

```
-- Move file: temp/ →    |
```

```
uploads/{table}/
```

```
-- DB::table()->insert()  |
```

```
-- afterInsert() hook
```

```
-- Find created record    |
```

```
<-- {success, data}-----
```

POST	/login	null	DoUnifiedLogin	Unified login (all roles)
------	--------	------	----------------	---------------------------

GET	/{model}/list		Get	view-{model}
-----	---------------	--	-----	--------------

GET	/doctor/appointments	doctor	DoctorAppointments
-----	----------------------	--------	--------------------

GET	/doctor/me	doctor	Doctor profile
-----	------------	--------	----------------

POST	/upload-tmp		api
------	-------------	--	-----

GET	/file/{path}	Serve file from document disk		
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Users	id, fullname, username, email, role_id, active, created_at, updated_at			
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roles	2	Super Admin (id=1), Staff
tasks	45	view-/show-/create-/update-/delete- for 9 entities
role_task	45	role_id=1 gets ALL tasks
rooms	5	Examination rooms, ICU, etc.
doctors	10	Various specializations
patients	25	With demographics
appointments	32	Mix of scheduled/completed/cancelled
medicalrecords	11	Linked to completed appointments

Appendix C: Environment Variables

```

JWT_SECRET=          # REQUIRED – not in .env.example, must add manually
DB_CONNECTION=mysql
DB_DATABASE=clinic_laravel
QUEUE_CONNECTION=database
SESSION_DRIVER=database
  
```

Appendix D: Key Dependencies

Backend (composer.json)

- laravel/framework ^13.8
- tymon/jwt-auth ^2.3
- laravel/sanctum ^4.0

Frontend (package.json)

- vue ^3.5
- pinia ^2
- vue-router ^4
- axios ^1
- tailwindcss ^4.0
- vite ^8.0

End of Reference Manual